Setup of a Microsoft DNSSEC-validating Resolver

1. Install Windows Server virtual machine

Microsoft DNS resolver is a build-in feature of the Windows Server. The ISO for a Windows Server virtual machine (VM) can be downloaded from Microsoft Evaluation Center. Note that this ISO is mainly intended for evaluation purposes, where the Windows OS is limited to a 180-day free trial period.

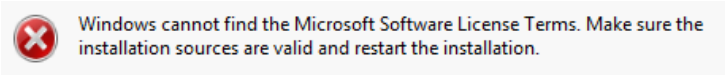
Before creating the VM, please make sure that:

- ✓ Your host machine is based on **x86-64 (AMD64)** architecture. The ISO of the Windows Server is not compatible with ARM-based devices, e.g., Apple Silicon (M-series) Macs.
- ✓ There is a **VM tool** (e.g., VMware Workstation) installed on your host machine.

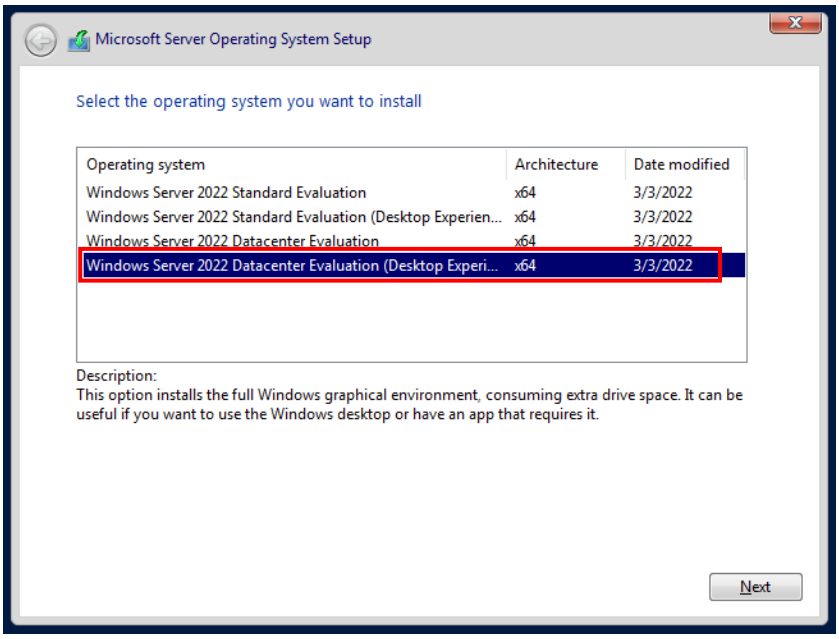
Import the ISO into the VM tool. Recommended configurations for the VM:

Version	# CPU core	RAM	Storage	Firmware	Network
Windows Server 2022 Datacenter	⩾2	⩾4GB	⩾60GB	BIOS	NAT

Please note that there is no need to enter the license key during the VM setup. If you encounter the following error when starting up the VM, please remove *Floppy* in the VM Settings menu, and restart the machine.



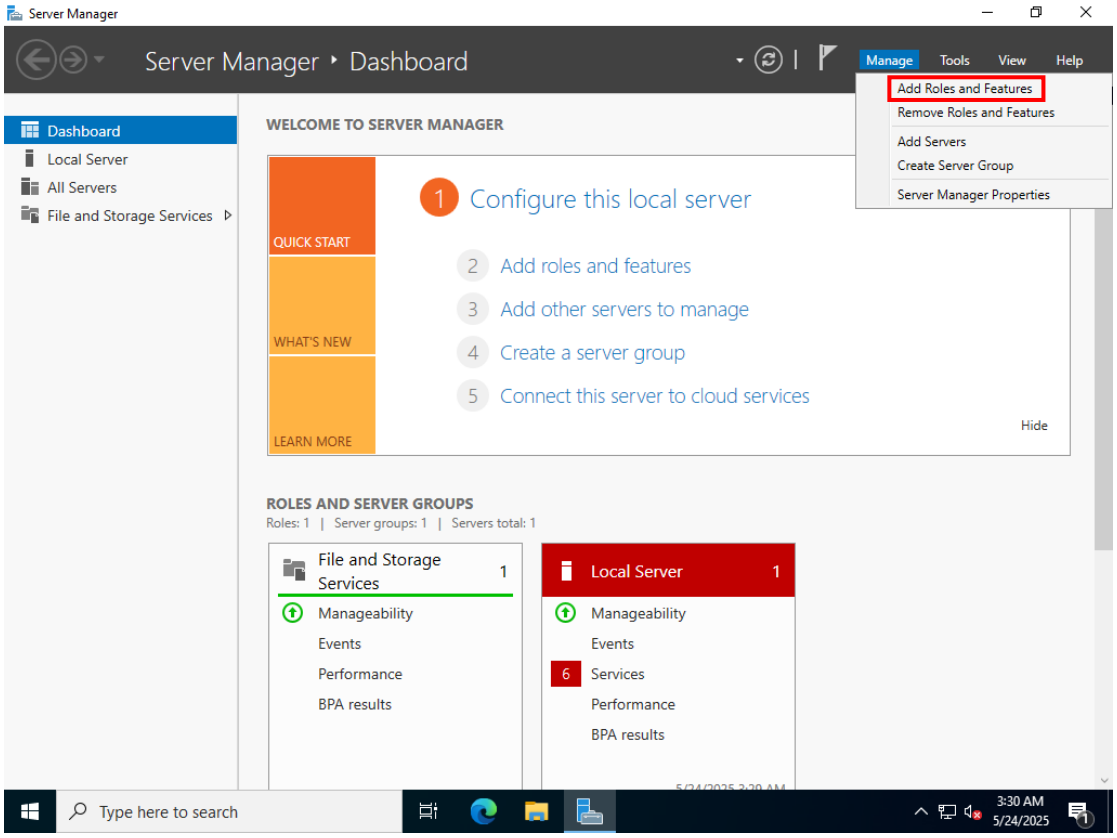
Select the OS with Desktop UI to install:



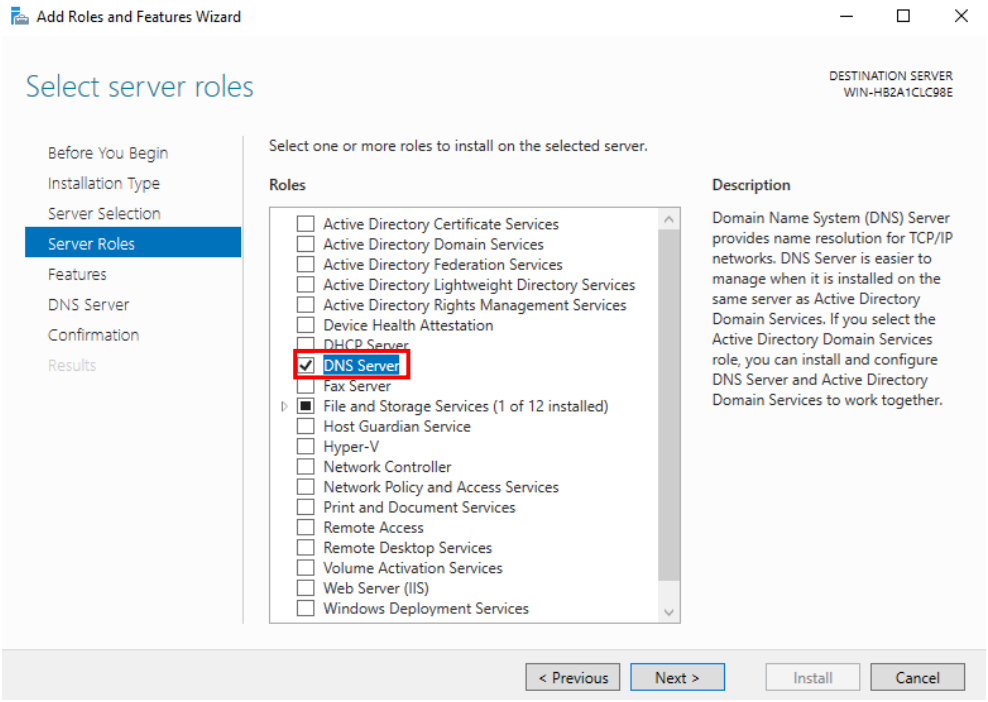
Then set your password and log in. Now the Windows Server VM is ready.

2. Add DNS service on Windows Server

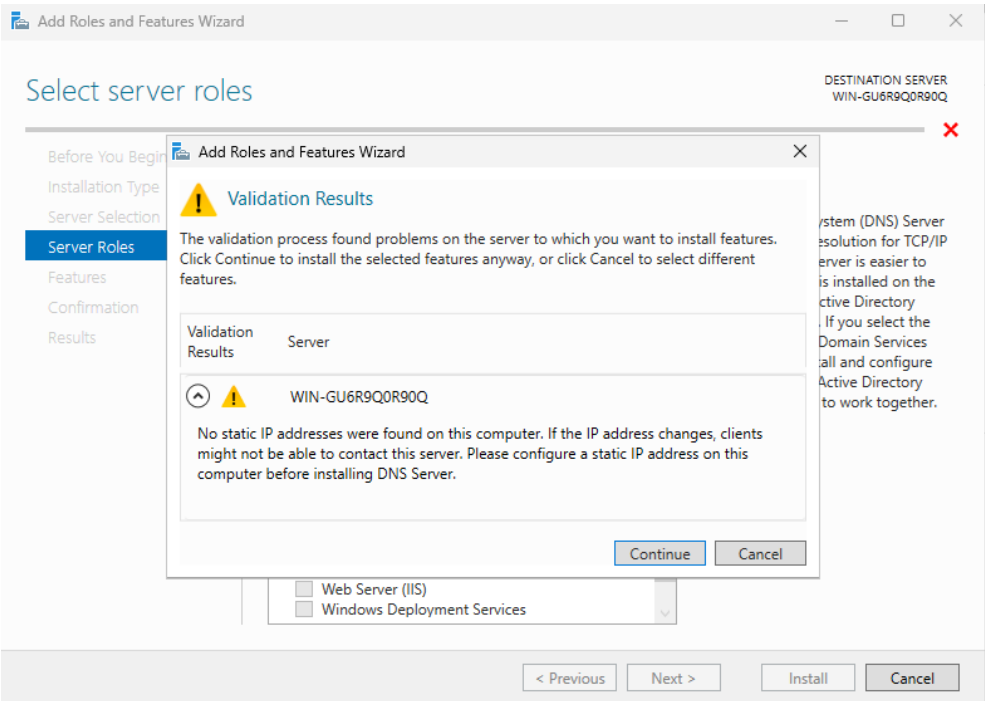
In Server Manager, select Manage -> Add Roles and Features.



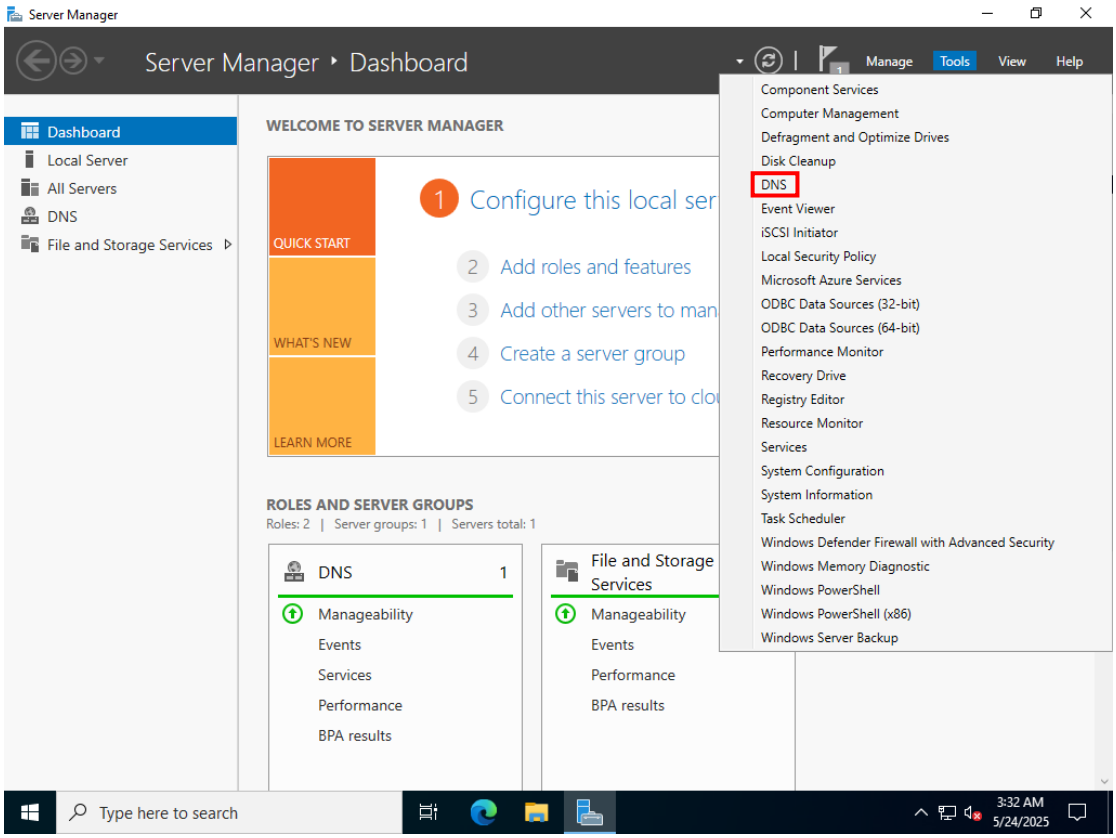
Click “Next”, until you see the list of server roles. Select “DNS Server”.



If you encounter the following warning, just click “Continue”. As all the experiments are conducted within the local network, the server does not need a static IP address for public access.



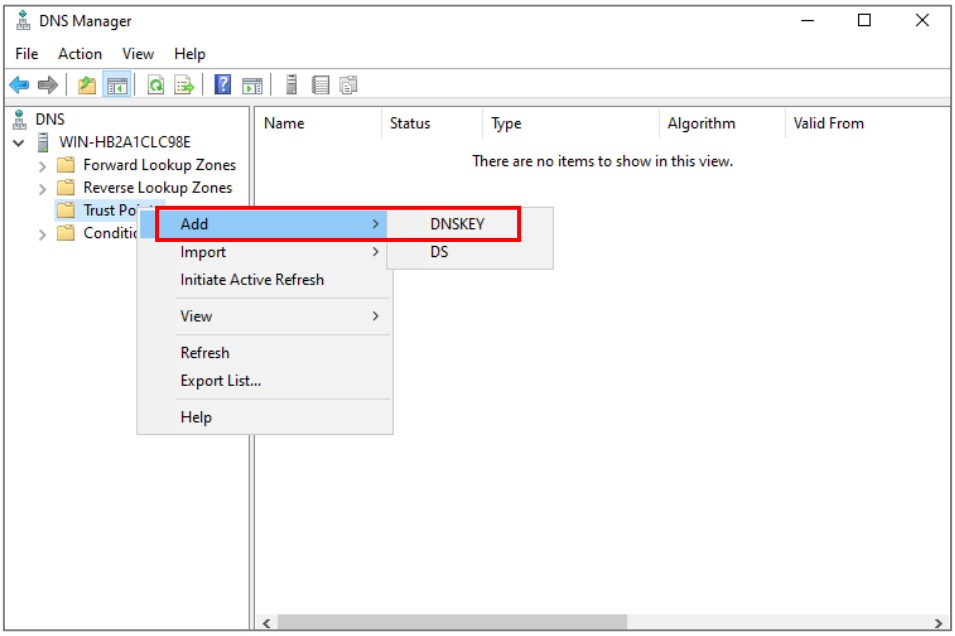
After installation, you can see “DNS” from the Tools menu.



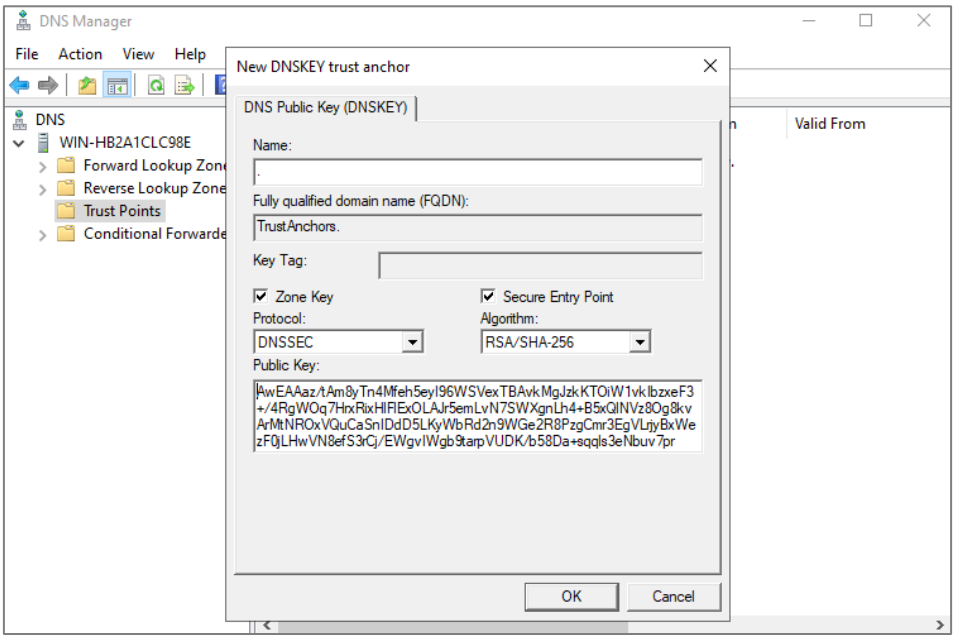
3. Enable DNSSEC validation

To enable DNSSEC validation on Microsoft DNS resolver, you should first import the DNSSEC trust anchor (i.e., DNSKEY of the DNS root) to the resolver.

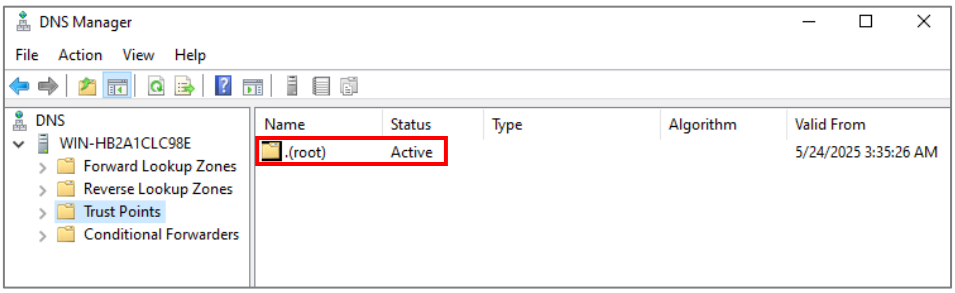
Click “DNS” in the Tools menu of the Server Manager, and you can see the newly-installed DNS server. Right-click “Trust Points”, select “Add” -> “DNSKEY”.



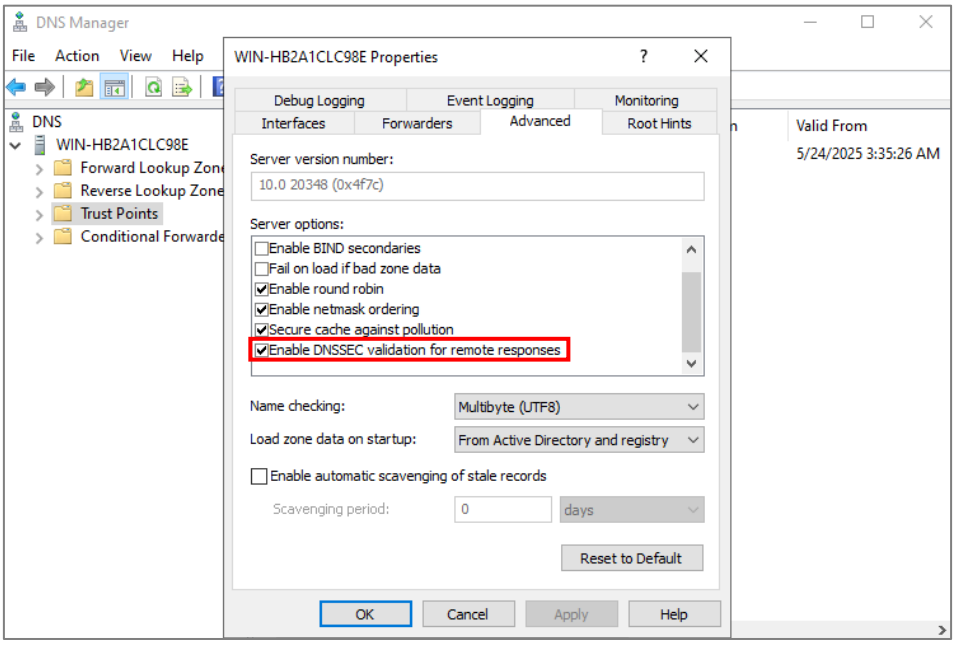
Enter “.” (the root zone) in the “Name” field. Retrieve the latest DNSSEC trust anchor (i.e., the KSK of the root zone) from [IANA root anchor](https://data.iana.org/root-anchors/root-anchors.xml) (https://data.iana.org/root-anchors/root-anchors.xml). Copy and paste the KSK with tag 20326 (start with AwEAAaz/tAm8yTn…) into the “Public Key” field. Then click “OK”.



Now the newly-added trust point should be in “Active” status.



Also please make sure that the DNSSEC validation property has been enabled: right-click the DNS server identifier (e.g., “WIN-HB2A1CLC98E” in the above figure) -> “Properties” -> “Advanced”, see if “Enable DNSSEC validation for remote responses” has been checked.



Now the setup of a Microsoft DNSSEC-validating resolver has finished.